

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250278515

Kind Code

A1

Publication Date

September 04, 2025

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SYSTEMS AND METHODS FOR TOKENIZATION OF PERSONALLY IDENTIFIABLE INFORMATION (PII)

Abstract

Described herein is a data security system for enabling tokenized access to sensitive data, including a token provider configured to connect to a remote client computing device over a secure communication channel, and cause display, at the remote client computing device, of a token request user interface including a selection form listing sensitive data elements associated with a first data subject. The token provider is also configured to receive a request for an access token, including a user selection of a subset of the sensitive data elements and one or more access authorization parameters, and generate an access token that enables access to only the subset of the sensitive data elements according to the authorization parameters. The token provider also stores the access token in a token database with the one or more authorization parameters, and transmits, to the remote client computing device, a response including the access token.

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Family ID: 79688309

Appl. No.: 19/199068

Filed: May 05, 2025

Related U.S. Application Data

parent US continuation 18190870 20230327 parent-grant-document US 12292995 child US 19199068

parent US continuation 16936136 20200722 parent-grant-document US 11615206 child US 18190870

Publication Classification

Int. Cl.: G06F21/62 (20130101); G06F16/23 (20190101); H04L9/40 (20220101)

U.S. Cl.:

CPC G06F21/6245 (20130101); G06F16/2379 (20190101); H04L63/10 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of U.S. patent application Ser. No. 18/190,870, filed Mar. 27, 2023, which is a continuation application of U.S. patent application Ser. No. 16/936,136, filed Jul. 22, 2020, now U.S. Pat. No. 11,615,206, which are hereby incorporated by reference herein in their entirety.

BACKGROUND

[0002] This disclosure relates generally to the field of data security and, more specifically, to the tokenization of personally identifiable information (PII).

[0003] There exist many situations in which individuals need to provide sensitive data, such as PII, to service providers, such as customer service representatives, bankers, health care providers, insurance claim adjusters, and the like. However, providing this sensitive data can be tedious and redundant. Moreover, many individuals may be uncomfortable providing their sensitive data, depending on the environment. For example, an individual in a public setting may not feel comfortable providing a credit card number or Social Security number to a service provider over the phone.

[0004] Therefore, there is a need for a system that minimizes redundancies in sharing sensitive data with trusted service providers and that maintains the security of the sensitive data.

BRIEF DESCRIPTION

[0005] In one aspect, a data security system for enabling tokenized access to sensitive data is provided. The data security system includes a token provisioning computing device including a processor communicatively coupled to a memory device. The token provisioning computing device is configured to receive, from a service provider computing device, a request for access to sensitive data of an individual stored in memory, and transmit an authorization request message to a remote client computing device associated with the individual, the authorization request message prompting the individual to authorize provisioning of an access token to the service provider, the access token being configurable by (i) associating a data definition of one or more sensitive data elements to the access token for allowing access to be granted, and (ii) identifying one or more authorization parameters. The token provisioning computing device is also configured to receive, from the remote computing device, an authorization response including the data definition and the one or more authorization parameters, generate the access token that enables access to the one or more sensitive data elements according to the one or more authorization parameters, and transmit, to the service provider computing device, a response including the access token.

[0006] In another aspect, a computer-implemented method for enabling tokenized access to sensitive data is provided. The method is implemented using a data security system including a token provisioning computing device including a processor communicatively coupled to a memory device. The method includes receiving, from a service provider computing device, a request for access to sensitive data of an individual stored in memory, and transmitting an authorization request message to a remote client computing device associated with the individual, the authorization request message prompting the individual to authorize provisioning of an access token to the service provider, the access token being configurable by (i) associating a data definition of one or more sensitive data elements to the access token for allowing access to be

granted, and (ii) identifying one or more authorization parameters. The method also includes receiving, from the remote computing device, an authorization response including the data definition and the one or more authorization parameters, generating the access token that enables access to the one or more sensitive data elements according to the one or more authorization parameters, and transmitting, to the service provider computing device, a response including the access token.

[0007] In a further aspect, a non-transitory computer-readable storage medium having computer-executable instructions stored thereon is provided. When executed by a processor of a token provisioning computing device of a data security computing system, the computer-executable instructions cause the processor to receive, from a service provider computing device, a request for access to sensitive data of an individual stored in memory, and transmit an authorization request message to a remote client computing device associated with the individual, the authorization request message prompting the individual to authorize provisioning of an access token to the service provider, the access token being configurable by (i) associating a data definition of one or more sensitive data elements to the access token for allowing access to be granted, and (ii) identifying one or more authorization parameters. The computer-executable instructions also cause the processor to receive, from the remote computing device, an authorization response including the data definition and the one or more authorization parameters, generate the access token that enables access to the one or more sensitive data elements according to the one or more authorization parameters, and transmit, to the service provider computing device, a response including the access token.

[0008] In yet another aspect, a data security system for enabling tokenized access to sensitive data is provided. The data security system includes a token provisioning computing device including a processor communicatively coupled to a memory device. The processor is programmed to communicatively connect to a remote client computing device over a secure communication channel, the remote client computing device operated by a first data subject, and to cause display, at the remote client computing device, of a token request user interface including a selection form listing a plurality of sensitive data elements associated with the first data subject. The processor is also programmed to receive, from the remote client computing device, a request for an access token, the request including a user selection of a subset of the sensitive data elements selected by the first data subject on the token request user interface and one or more access authorization parameters. The processor is further programmed to generate an access token that enables access to only the subset of the sensitive data elements according to the one or more authorization parameters, store the access token in a token database with the one or more authorization parameters, and transmit, to the remote client computing device, a response including the access token.

[0009] In another aspect, a computer-implemented method for enabling tokenized access to sensitive data is provided. The method is implemented using a data security system including a token provisioning computing device including a processor communicatively coupled to a memory device. The method includes communicatively connecting to a remote client computing device over a secure communication channel, the remote client computing device operated by a first data subject, and causing display, at the remote client computing device, of a token request user interface including a selection form listing a plurality of sensitive data elements associated with the first data subject. The method also includes receiving, from the remote client computing device, a request for an access token, the request including a user selection of a subset of the sensitive data elements selected by the first data subject on the token request user interface and one or more access authorization parameters, and generating an access token that enables access to only the subset of the sensitive data elements according to the one or more authorization parameters. The method further includes storing the access token in a token database with the one or more authorization parameters, and transmitting, to the remote client computing device, a response

including the access token.

[0010] In a further aspect, a non-transitory computer-readable storage medium having computer-executable instructions stored thereon is provided. When executed by a processor of a token provisioning computing device of a data security computing system, the computer-executable instructions cause the processor to communicatively connect to a remote client computing device over a secure communication channel, the remote client computing device operated by a first data subject. The computer-executable instructions also cause the processor to cause display, at the remote client computing device, of a token request user interface including a selection form listing a plurality of sensitive data elements associated with the first data subject, and receive, from the remote client computing device, a request for an access token. The request includes a user selection of a subset of the sensitive data elements selected by the first data subject on the token request user interface and one or more access authorization parameters. The computer-executable instructions further cause the processor to generate an access token that enables access to only the subset of the sensitive data elements according to the one or more authorization parameters, store the access token in a token database with the one or more authorization parameters, and transmit, to the remote client computing device, a response including the access token.

[0011] In yet another aspect, a data security system for enabling tokenized access to sensitive data is provided. The data security system includes a token provisioning computing device including a processor communicatively coupled to a memory device. The token provisioning computing device is configured to initiate a secure connection with a remote client computing device of a first data subject, and receive, from the remote client computing device of the first data subject, a request for an access token to provide a service provider computing device with access to sensitive data associated with the first data subject. The request includes a data definition of the sensitive data to which access is to be provided and one or more authorization parameters. The token provisioning computing device is also configured to generate the access token that enables access to the defined sensitive data according to the one or more authorization parameters, store the access token in a token database with the data definition and the one or more authorization parameters, and transmit, to the remote client computing device of the first data subject, a response including the access token and instructions that enable the remote computing device to at least one of display the access token to the first data subject and transmit the access token to the service provider computing device.

[0012] In another aspect, a computer-implemented method for enabling tokenized access to sensitive data is provided. The method is implemented using a data security system including a token provisioning computing device including a processor communicatively coupled to a memory device. The method includes initiating, by the token provisioning computing device, a secure connection with a remote client computing device of a first data subject, and receiving, by the token provisioning computing device from the remote client computing device of the first data subject, a request for an access token to provide a service provider computing device with access to sensitive data associated with the first data subject. The request includes a data definition of the sensitive data to which access is to be provided and one or more authorization parameters i. The method also includes generating, by the token provisioning computing device, the access token that enables access to the defined sensitive data according to the one or more authorization parameters, storing, by the token provisioning computing device, the access token in a token database with the data definition and the one or more authorization parameters, and transmitting, by the token provisioning computing device to the remote client computing device of the first data subject, a response including the access token and instructions that enable the remote computing device to at least one of display the access token to the first data subject and transmit the access token to the service provider computing device.

[0013] In a further aspect, a non-transitory computer-readable storage medium having computer-executable instructions stored thereon is provided. When executed by a processor of a token

provisioning computing device of a data security computing system, the computer-executable instructions cause the processor to initiate a secure connection with a remote client computing device of a first data subject, and receive, from the remote client computing device of the first data subject, a request for an access token to provide a service provider computing device with access to sensitive data associated with the first data subject. The request includes a data definition of the sensitive data to which access is to be provided and one or more authorization parameters. The computer-executable instructions also cause the processor to generate the access token that enables access to the defined sensitive data according to the one or more authorization parameters, store the access token in a token database with the data definition and the one or more authorization parameters, and transmit, to the remote client computing device of the first data subject, a response including the access token and instructions that enable the remote computing device to at least one of display the access token to the first data subject and transmit the access token to the service provider computing device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIGS. 1-7 show example embodiments of the methods and systems described herein.

[0015] FIG. 1 is a schematic diagram illustrating a first example data security system for enabling tokenized access to sensitive data, in accordance with the present disclosure.

[0016] FIG. 2 illustrates an example user interface displayed on a client computing device of the data security computing system shown in FIG. 1, including generation of a data definition.

[0017] FIGS. 3 and 4 are swim lane diagrams illustrating implementation of a tokenized data access method using components of the data security computing system shown in FIG. 1.

[0018] FIG. 5 illustrates an example client computing device that may be used with the data security computing system shown in FIG. 1.

[0019] FIG. 6 illustrates an example server computing device that may be used with the data security computing system shown in FIG. 1.

[0020] FIG. 7 illustrates an example tokenized data access method that may be implemented using the data security computing system shown in FIG. 1.

[0021] Like numbers in the Figures indicate the same or functionally similar components.

DETAILED DESCRIPTION

[0022] The present disclosure relates to a data security computing system that enables tokenized access to sensitive data, such as personally identifiable information (PII). In particular, rather than providing data directly to a service provider, an individual (also referred to herein as a “data subject” or “token requestor”), who is also a subject and/or an originating source of the PII, may request a token, to be securely provided to the service provider, that enables the service provider to securely access stored data associated with that individual or data subject. This process eliminates redundancies while enhancing data security as well as data accuracy. Specifically, the individual need not repetitively provide the same data elements to many different services, which is tedious and time-consuming but also vulnerable to user error (either by the individual or the service provider). The service provider uses the token to access data that is both securely stored and accurately transmitted (and, in some embodiments, accurately locally entered at the service provider).

[0023] In the example embodiment, a data subject (e.g., an individual) registers with the data security computing system and provides their PII for secure storage. The data subject provides their PII as one or more data elements, for example, using a client computing device. The client computing device is communicatively coupled to a centralized server computing device referred to herein as a “secure manager.” The secure manager is configured to store PII in one or more

centralized or decentralized databases. The secure manager stores the PII in encrypted or otherwise secured format within the database(s). As described further herein, the secure manager is configured to limit access to the data stored within the databases, specifically, to access defined by an access token requested and provided by the data subject to a service provider.

[0024] In one embodiment, the client computing device is communicatively coupled to the secure manager via a secure communication channel that is initiated in a web browser or software application (“app”) environment. For example, the client computing device stores and executes an app that initiates a secure communication channel with the secure manager. The app may cause display of one or more screens on the client computing device, including data entry and/or data provision screens. To provide their PII for storage, the data subject may manually enter data elements into their client computing device, such as via a manually fillable form. Alternatively, the client computing device may automatically populate certain fields or data elements using stored information (e.g., name, date of birth, address, etc.). The data subject controls which data elements to provide for storage to the secure manager, and may choose to populate fewer than all available fields. The data subject transmits the PII as one or more data elements to the secure manager for secure storage in one or more database(s). In some embodiments, PII associated with the data subject is provided to the secure manager by other data sources (e.g., insurance companies may provide claim details/history, banks may provide loan information, etc.). The secure manager may additionally or alternatively store a pointer to PII that is stored in other locations (e.g., other than the database of the secure managers).

[0025] The secure manager indexes the stored PII according to one or more variables, such as a subject identifier, or any other variable that uniquely identifies the data subject. The subject identifier may be provided by the data subject (e.g., as a phone number or SSN), or may be automatically generated by the secure manager during the registration phase (e.g., as a pseudo-random alphanumeric code). In such cases, the secure manager returns the subject identifier to the client computing device for storage (e.g., within the secure app environment). The client computing device may include the subject identifier in further communications with service providers and/or the secure manager, such that the data subject's PII is easily retrieved using the subject identifier. The data subject may use their client computing device to add, delete, and/or modify their stored PII at any time.

[0026] The data subject (e.g., an individual) may wish to provide access to their stored PII at a later date, to a service provider. A service provider may include, for example, an insurance claims manager, a customer service representative, a health care provider, and the like. The service provider requests various data elements of PII from the data subject. For example, the service provider may ask the data subject to verbally provide PII, to physically write down their PII on one or more forms, or to provide PII in electronic format (e.g., in an electronic form). The data subject may not wish to provide that PII for one or more reasons. For example, the data subject may be in a public place where verbalizing their PII may make the PII vulnerable to being overheard. Additionally or alternatively, the data subject may merely not wish to provide their data in such a tedious and redundant fashion.

[0027] Accordingly, to maintain security of their PII and/or to avoid physically or manually providing this data, the data subject requests an access token, which the service provider can use to securely access at least a portion of the data subject's stored PII. In the example embodiment, the data subject uses their client computing device to request such an access token. The data subject initiates a secure communication channel with a token provider within a web browser and/or app environment. The token provider is configured to generate tokens that enable data subject-defined access to the data subject's PII. In some embodiments, the token provider is the same entity or is otherwise associated with the secure manager. That is, one (business) entity may provide data storage services as well as token provisioning services. Notably, in such embodiments, the token provider is an independent computing device or is an independent processing module of the same

computing device, such that the token provision services remain independent of the data storage services. In other embodiments, the token provider is wholly independent of any data storage providers.

[0028] To request the access token, the data subject (e.g., the individual) selects which data elements of PII to allow the service provider access to. The client computing device displays a list of all available PII. The list may be dynamically populated by the secure manager. For example, the app environment may enable an API connection between the secure manager, such that the real-time currently available PII is shown in the list. Alternatively, the list may be populated by the client computing device based upon data that is periodically (e.g., daily, weekly, monthly) provided by the secure manager, reflecting the data elements available as of the last periodic update.

[0029] The data subject selects one or more data elements of PII to allow the service provider access (e.g., using checkboxes, radio buttons, a drag-and-drop interface, etc.). The selected data may be referred to herein as a “data definition,” which defines which data elements the service provider will be allowed access to. The data subject may also select one or more authorization parameters that further define how the service provider is allowed to access the data subject's PII. The authorization parameters may include an authorization date or validity date parameter. This parameter defines how long the service provider is allowed access to and/or local storage of the data subject's PII. For example, where a data subject is providing PII to an insurance claim adjuster, the data subject may authorize the insurance adjuster to access their PII for a period corresponding to the claim processing timeline, such as three months or six months. As another example, where a data subject is providing PII (e.g., an SSN) to a customer service provider (e.g., for authentication to a bank), the data subject may only authorize access for five or ten minutes.

[0030] The authorization parameters may include one or more authorized service providers. For example, where a data subject is having their vehicle serviced as a result of an automobile accident, in accordance with an insurance claim, the data subject may only authorize one specific service provider location (e.g., repair shop) in a nationwide network of such service providers. The authorization parameters may include additional or alternative parameters, such a data access source and/or a date range of accessible data. For example, a data subject may authorize access to PII within a date range of the last five or ten years, but not any PII from dates outside of that range.

[0031] The data subject, after making all necessary and/or desired selections, requests an access token. The client computing device transmits an access token request to the token provider. The request includes the data definition and any associated authorization parameters. In some embodiments, the request also includes the subject identifier (e.g., a phone number, SSN, other unique identifier). The request may include still other data elements, such as authentication data elements that may be used to authenticate the request and/or the data subject. For example, the data subject may be required to enter log-in and/or authentication credentials to open their app and/or initiate the secure connection with the token provider. In such cases, the client computing device may transmit an indicator, in the token request message, that indicates the data subject successfully logged-in and/or authenticated themselves to the app prior to generating the token request.

[0032] The token provider receives the request from the client computing device and processes the request to generate an access token. The token provider transmits the access token back to the client computing device, such that the data subject may provide the access token to the service provider. The access token is associated with and/or includes the data definition and authorization parameters, and is specific to this particular instance in which the data subject desires to give this particular service provider access to the selected PII. That is, in the example embodiment, the access token is not usable in any other instance. In an alternative embodiment, the access token may be authorized for use by that particular service provider in multiple instances, to access the same selected PII-however, such repeated access may require specific authorization by the data subject (e.g., via an authorization parameter that enables repeat usage of the access token, or by requesting confirmation from the data subject if/when a service provider attempts to re-use an

access token).

[0033] In the example embodiment, the access token is embodied as an alphanumeric code, and the data subject provides the access token to the service provider verbally, shows the access token to the service provider on their client computing device, or provides the access token electronically to the service provider (e.g., by typing or pasting the code into a fillable field). In some alternative embodiments, the access token may be otherwise embodied, such as a string or message transmittable over near-field communication (NFC, e.g., by “tapping” the client computing device at a receiver of the service provider), or as image content (e.g., a bar code or QR code that may be scanned by the service provider). In some embodiments, the client computing device also provides the subject identifier to the service provider.

[0034] In some embodiments, the token provider authenticates the data subject before generating and/or providing the requested access token. In some such embodiments, the token provider uses authentication elements provided in the token request to authenticate the data subject. In some embodiments, the token provider transmits an authentication request message back to the client computing device, which includes instructions that cause the client computing device to prompt the data subject to provide authentication credentials (e.g., a biometric identifier, a password, a PIN, answer(s) to security question(s), etc.). In these embodiments, the token provider receives an authentication response message back from the client computing device including the data subject's responses, and processes those responses (e.g., by comparing the data subject's input to stored authentication credentials) to authenticate the data subject.

[0035] It is contemplated that, in some embodiments, the service provider may request an access token from the token provider, on behalf of the data subject. In such embodiments, the token provider, upon receiving a token request message from the service provider, will request confirmation, a data definition, and/or authorization parameters from the data subject (e.g., by transmitting a confirmation request message to the client computing device associated with the data subject). When the data subject confirms the request, and provides any associated data definition and/or authorization parameters, the token provider generates the access token, as described above, and transmits the access token directly to the service provider (e.g., in a token response message).

[0036] The token provider stores the access token in a database. The token provider may also store the data definition and/or the authorization parameters along with the access token. The token provider allows the data subject to access any generated access token(s) via their client computing device. The data subject may review which access tokens have been generated, which service provider(s) have access to which data element(s), how long authorization was granted, and the like. In some embodiment, the data subject is able to select an access token for modification or revocation. The data subject may add or remove data element(s) they wish to provide access to, and/or may modify the authorization parameters (e.g., increase or decrease the period of data validity). The data subject may revoke any access token. In response to the data subject modifying or revoking an access token, the token provider may be configured to generate a message indicating the modification/revocation, for transmission to the associated service provider (either directly, from the token provider, or indirectly, via the client computing device). In response to the data subject revoking an access token, the token provider may delete the stored access token and/or disable the access token, which prevents further access to the data subject's data by the service provider. Additionally, where an access token has an associated authorization parameter that specifies a validity date, when the validity date is reached (or passes), the token provider may delete the stored access token and/or disable the access token, which prevents further access to the data subject's data by the service provider.

[0037] The service provider receives the access token from the data subject, as described above. The service provider provides the access token to an interface executed on a service provider computing device, where the interface is maintained by the secure manager in one example embodiment. The service provider may be a human that enters the access token, for example, via a

keyboard, mouse, and the like. The service provider may additionally or alternatively include an input device, such as an NFC receiver, scanner, or computing device that receives the access token electronically. The secure manager receives the access token from the service provider computing device.

[0038] The data storage provider initiates a secure communication channel with the token provider. Where the secure manager and token provider are a same entity, the secure communication channel may include communicative coupling between independent computing devices or processing modules. Where the secure manager and token provider are not a same entity, the secure communication channel may be any secure communication channel between remote computing devices.

[0039] As described in greater detail herein, the secure manager exchanges messages with the token provider to validate the access token received from the service provider. The data storage provider sends a validation request message to the token provider, the validation request message including the access token and, in some embodiments, a subject identifier. The subject identifier may be the same as the subject provider transmitted from the client computing device, or may be a separate subject identifier. The token provider compares the received access token (and, in some embodiments, the subject identifier) to the stored access token to determine whether the received access token is valid. When the access token is invalid, the token provider returns a validation response including an access denial, and an indication that the access token is invalid. When the access token is valid, the token provider returns a validation response including an access approval. In some embodiments, the access approval may include the data definition and/or the authorization parameters associated with the valid stored access token. In some embodiments, the access token itself defines what data is accessible.

[0040] When the secure manager receives the validation response including the access approval (and, in some embodiments, the data definition and/or authorization parameters), the data storage provider provides access to PII under the conditions of the access token. That is, the data storage provider transmits the approved and authorized data back to the service provider computing device. Therefore, the service provider computing device receives (and may locally store) the PII without the data subject having to provide the PII, especially where the conditions of such provision may make the PII vulnerable to unauthorized access.

[0041] As used herein, personally identifiable information (PII) refers to information that, when used alone or in combination, can identify a specific individual. PII may include direct identifiers (e.g., name, Social Security Number (SSN), phone number, etc.) or indirect identifiers, also known as quasi-identifiers, that can be combined together to identify the individual (e.g., date of birth, address, etc.). PII may include, but is not limited to, name, address, phone number, SSN, date of birth, educational history, social media identifiers, IP address or other device/network identifiers, driver's license number, passport number, payment account/device (e.g., credit card) information, other financial information, and the like.

[0042] The data security system described herein improves over conventional methods of providing sensitive data, including PII, to service providers. Specifically, the data is secured and not vulnerable to unauthorized access (e.g., by overhearing, by written or printed details being misplaced or copied, etc.). This data security system reduces data subject input requirements to a few selections and commands, and centralizes storage and retrieval to secured computing devices and databases. Moreover, this data security system improves data accuracy across multiple devices by ensuring a singular and/or unified data source is providing PII to all requesting service providers.

[0043] The technical problems addressed by this system include at least one of: (i) vulnerability of PII to unauthorized access when transmitted verbally or on paper to service providers; (ii) lack of centralized PII data sources and access points; (iii) discrepancies in data between service providers due to user entry error and/or varied system formats/interfaces; and (iv) time-consuming,

inefficient, and redundant provision of PII to various service providers (e.g., manual and/or verbal entry).

[0044] The methods and systems described herein may be implemented using computer programming or engineering techniques including computer software, firmware, hardware, or any combination or subset thereof, wherein the technical effects may be achieved by: (a) initiating, by the token provisioning computing device, a secure connection with a remote client computing device of a first data subject; (b) receiving, by the token provisioning computing device from the remote client computing device of the first data subject, a request for an access token to provide a service provider computing device with access to sensitive data associated with the first data subject, wherein the request includes a data definition of the sensitive data to which access is to be provided and one or more authorization parameters; (c) generating, by the token provisioning computing device, the access token that enables access to the defined sensitive data according to the one or more authorization parameters; (d) storing, by the token provisioning computing device, the access token in a token database with the data definition and the one or more authorization parameters; and/or (e) transmitting, by the token provisioning computing device to the remote client computing device of the first data subject, a response including the access token and instructions that enable the remote computing device to at least one of display the access token to the first data subject and transmit the access token to the service provider computing device.

[0045] The resulting technical benefits achieved by this system include at least one of: (i) providing a secure, centralized access point for PII; (ii) unification and standardization of PII provided to multiple service providers across disparate systems; and (iii) securing the transmission of PII between data subjects and service providers while reducing user effort and time required.

[0046] As used herein, a processor may include any programmable system including systems using micro-controllers, reduced instruction set circuits (RISC), application specific integrated circuits (ASICs), logic circuits, and any other circuit or processor capable of executing the functions described herein. The above examples are example only, and are thus not intended to limit in any way the definition and/or meaning of the term “processor.”

[0047] As used herein, the terms “software” and “firmware” are interchangeable, and include any computer program stored in memory for execution by a processor, including RAM memory, ROM memory, EPROM memory, EEPROM memory, and non-volatile RAM (NVRAM) memory. The above memory types are example only, and are thus not limiting as to the types of memory usable for storage of a computer program.

[0048] In one embodiment, a computer program is provided, and the program is embodied on a computer readable storage medium. In an example embodiment, the system is executed on a single computer system, without requiring a connection to a server computer. In a further embodiment, the system is being run in a Windows® environment (Windows is a registered trademark of Microsoft Corporation, Redmond, Washington). In yet another embodiment, the system is run on a mainframe environment and a UNIX® server environment (UNIX is a registered trademark of X/Open Company Limited located in Reading, Berkshire, United Kingdom). The application is flexible and designed to run in various different environments without compromising any major functionality. In some embodiments, the system includes multiple components distributed among a plurality of computing devices. One or more components may be in the form of computer-executable instructions embodied in a computer-readable medium. The systems and processes are not limited to the specific embodiments described herein. In addition, components of each system and each process can be practiced independent and separate from other components and processes described herein. Each component and process can also be used in combination with other assembly packages and processes.

[0049] As used herein, an element or step recited in the singular and proceeded with the word “a” or “an” should be understood as not excluding plural elements or steps, unless such exclusion is explicitly recited. Furthermore, references to “example embodiment” or “one embodiment” of the

present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

[0050] FIG. 1 includes a schematic diagram illustrating an example embodiment of a data security computing system **100** for enabling tokenized access to sensitive data, in accordance with the present disclosure. In the illustrated embodiment, data security computing system **100** includes a plurality of service providers **102**, at least one secure manager **104** communicatively coupled to one or more secure databases **106** for securing storing sensitive data (including PII), and at least one token provisioning computing device (“token provider”) **108**. Service providers **102** collectively include service provider computing devices (e.g., desktop computing devices, laptop computing devices, tablets, etc.) and human operators that may employ service provider computing devices (e.g., to enter data). Service providers **102** are communicatively coupled to secure managers **104** to transmit access tokens thereto and to request and receive sensitive data therefrom. In addition, secure managers **104** are communicatively coupled to token provider **108**, to validate received access tokens before providing sensitive data to service providers **102**. In some embodiments, service providers **102** are communicatively coupled to token provider **108**, and may request/receive access tokens directly therefrom.

[0051] In addition, client computing devices **110**, operated by individuals or data subjects **112**, are communicatively coupled to token provider **108**. Data subject **112** operates client computing device **110** to request access tokens from token provider **108** over a secure communication channel. Data subject **112** operates client computing device **110** to select data elements to provide access to, as well as authorization parameters of that access, as described herein. Data subject **112** also uses client computing device **110** to receive access tokens for provision to service provider **102** (via data subject **112**, such as verbally or in written form, or directly, such as via NFC communication). Client computing devices **110** are also communicatively coupled to secure manager **104**, such that data subject **112** may provide PII and/or PHI for storage, by secure manager **104**, within database(s) **106**.

[0052] Secure manager **104** may include any suitable computing device(s), such as one or more server computing device(s), cloud-based computing and/or storage systems, and/or any other device(s). Secure manager **104** stores sensitive data in databases **106**, which are any suitable secure storage devices, including any suitable encryption and/or access restriction mechanisms.

[0053] Likewise, token provider **108** may include any suitable computing device(s), such as one or more server computing device(s), databases, cloud-based computing and/or storage systems, and/or any other device(s), token databases/vaults, and/or encryption mechanisms. As described above, token provider **108** may be a part of and/or associated with secure manager **104**. Alternatively, token provider **108** is independent from secure manager **104**.

[0054] Client computing device **110** may include any device capable of accessing the Internet and communicating with service provider **102**, secure manager **104**, and/or token provider **108**, including a laptop computing device, mobile phone (e.g., smartphone), tablet, desktop computing device, and the like. Moreover, data subject **112** may operate more than one client computing device **110** to perform the processes described herein. For example, data subject **112** may operate a desktop client computing device **110** to provide PII for storage by secure manager **104**, and may operate a smartphone client computing device **110** to request an access token from token provider **108**, receive the access token from token provider, and/or provide the access token to service provider **102**.

[0055] FIG. 2 illustrates one example token request environment **200**, embodied as a website or user interface of a software app executed on client computing device **110** (shown in FIG. 1). In some embodiments, data subject **112** (also shown in FIG. 1) authenticates into the token request environment **200**. For example, data subject **112** may provide a user name and password, a biometric identifier (e.g., a fingerprint or facial image), a PIN, a one-time PIN, and/or any other authentication information.

[0056] Data subject **112** navigates to the particular interface **202** shown in FIG. 2 by selecting a command to generate an access token. For example, as described elsewhere herein, data subject **112** may wish to provide PII to a service provider **102** (shown in FIG. 1) without making their PII vulnerable to unauthorized access and without having to manually provide their PII (e.g., verbally, written, typed, etc.). As shown in interface **202**, data subject **112** has provided or selected a plurality of data elements **204** of PII that are available to provide to service provider(s) **102** using an access token.

[0057] Data subject **112** selects one or more data elements **204** that data subject **112** wishes to provide to service provider **102**. It should be understood that data subject **112** may not wish to provide all data elements **204** to a service provider **102**, and may therefore select fewer than all available data elements **204**. For example, it may not be necessary to provide a driver's license number to a service provider associated with a financial institution. In such an example, data subject **112** would not select those data elements **204**.

[0058] Data subject **112** may also include additional authorization parameters, including validity or access restrictions **206**, such as an authorization/validity date after which the service provider **102** is not authorized to access or store their data. Data subject **112** may select a specific date after which they wish to restrict or revoke access to their sensitive data, or may authorize such access indefinitely (e.g., until the data subject manually revokes an access token, or until a regulatory or statutory period of data retention expires). Although not shown, additional or alternative authorization parameters may be available for selection via interface **202**, such as a selection of authorized service providers and/or data sources.

[0059] When data subject **112** is satisfied with their selection(s), data subject **112** selects a generation control **208**, embodied here as a selectable button. Client computing device **110** generates an access token request message including a data definition of all selected data elements **204** and any authorization parameters selected and/or defined by data subject **112**. Client computing device **110** transmits the access token request to token provider **108** (shown in FIG. 1).

[0060] Although not shown in FIG. 2, token request environment **200** may display additional and/or alternative interfaces. In particular, token request environment **200** may display, to data subject **112**, an inventory of any/all access tokens that are active (e.g., still valid) and/or ever requested by the data subject **112**. Data subject **112** may select any such access tokens for modification (e.g., to expand or contract the data available to service provider **102** via the access token, to extend or reduce the validity of the access token, etc.) or revocation.

[0061] FIGS. 3 and 4 are swim lane diagrams illustrating an example tokenized data access process **300** implemented using components of data security computing system **100**, shown in FIG. 1. In the example embodiment, process **300** includes several stages, including a token generation stage **302** and a token usage stage **304**.

[0062] In token generation stage **302**, as shown in FIG. 3, a user (e.g., data subject **112**) logs in and/or authenticates into (**310**) a web browser and/or app environment on their client computing device **110**, which may be maintained by token provider **108**. As described herein, log-in/authentication (**310**) may include providing log-in/authentication credentials to client computing device **110**, such as a username/password, PIN, biometric identifier (e.g., password, facial recognition, retinal scan, etc.), and the like. Thereafter, data subject **112** is authorized to interact with token provider **108** (via client computing device **110**) to request and receive access tokens.

[0063] More specifically, data subject **112** selects information (**312**) that data subject **112** wishes to provide a service provider (e.g., a service provider **102**) access to, via an access token. This selection step (**312**) may be implemented using an interface **202** as shown and described with respect to FIG. 2, and/or using any other web browser and/or app interface. Client computing device **110** generates and transmits (**314**) an access token request message, including the data definition and authorization parameters selected by the data subject **112**, to token provider **108**. Token provider **108** generates the access token as described herein, such as an alphanumeric code

or computer-readable image content (e.g., a barcode or QR code).

[0064] Although not shown in FIG. 3, in some embodiments, token provider **108** authenticates data subject **112** prior to generating and/or providing the access token. For example, client computing device **110** may transmit (314) the log-in/authentication credentials entered by data subject **112** during the log-in/authentication step (310), or an indication of successful authentication, in the access token request message, and token provider **108** may use those credentials or indication to authenticate data subject **112**. Additionally or alternatively, token provider **108** transmits an authentication request message back to client computing device **110** with instructions for client computing device **110** to request additional/alternative authentication credentials from data subject **112**, which client computing device **110** returns to token provider **108** in an authentication response message.

[0065] Token provider **108** is configured to generate the access token according to data subject's data definition and authorization parameters, and to transmit (316) an access token response message, including the access token, back to client computing device **110**. Although not shown in FIG. 3, token provider **108** also stores the access token in a secure memory location (e.g., an integral memory, a centralized token database, a non-centralized token database, etc.).

[0066] Client computing device **110** receives the access token from token provider **108**. In some embodiments, client computing device **110** provides or displays (318) the access token to data subject **112**. For example, where the access token is an alphanumeric code, client computing device **110** displays (318) the access code on a user interface thereof. Data subject **112** views the access token and provides (319) the access token to service provider **102** (e.g., by reading the access token to service provider **102** aloud, writing the access token, typing or pasting the access code into a fillable field, etc.). Additionally or alternatively, client computing device **110** transmits (320) the access token directly to service provider **102** (e.g., via NFC "tap" or other NFC communication, via SMS/text, via email, etc.).

[0067] Once service provider **102** has received the access token, process **300** proceeds to token usage stage **304**, shown in FIG. 4. Service provider **102** transmits (322) the access token to secure manager **104**, or the source of the data that service provider **102** wishes to access. Service provider **102** transmits (322) the access token to secure manager **104** over any suitable communication channel. In some embodiments, service provider **102** also transmits a subject identifier, which may be received from data subject **112** (e.g., during provision 319) and/or from client computing device **110** (e.g., during transmission 420).

[0068] Secure manager **104** receives the access token, and transmits (324) the access token to token provider **108** in a token validation request message. Token provider **108** receives the access token from secure manager **104** and performs a lookup in the memory (e.g., database) to retrieve any matching stored access token. Where token provider **108** also receives a subject identifier, token provider **108** may use the subject identifier to perform the lookup operation, to retrieve active access tokens associated with data subject **112**. Token provider **108** compares the received access token to the stored/retrieved access token, to determine if the access token is valid (e.g., is received from an authorized service provider **102**, is within a validity date, is seeking data from the appropriate secure manager **104**, etc.). Token provider **108** transmits (326) a token validation response message back to secure manager **104**, the token validation response message indicating whether the access token was successfully validated. When the access token is successfully validated, the token validation response message may also include instructions that cause secure manager **104** to provide the requested data to service provider **102**.

[0069] When the access token is successfully validated, secure manager **104** retrieves (e.g., from database(s) **106**) the sensitive data (PII) that has been requested by service provider **102**. Secure manager **104** transmits (328) the data back to service provider **102**.

[0070] Although not shown in FIG. 4, in alternative embodiment, service provider **102** may communicate directly with token provider **108** to validate the access token prior to sending the

access token to secure manager **104**. In such embodiments, service provider **102** transmits the access token and/or an indication of successful validation of the access to secure manager **104**. In response, secure manager **104** retrieves and transmits the requested and authorized data back to service provider **102**.

[0071] FIG. 5 illustrates an example configuration of a user system **500**, such as client computing device **110** and/or a service provider computing device of service provider **102** (both shown in FIG. 1). In the example embodiment, user system **500** includes a processor **502** for executing instructions. In some embodiments, executable instructions are stored in a memory area **504**. Processor **502** may include one or more processing units, for example, a multi-core configuration. Memory area **504** is any device allowing information such as executable instructions to be stored and retrieved. Memory area **504** may include one or more computer readable media.

[0072] User system **500** also includes at least one media output component **506** for presenting information to a user **508** (e.g., data subject **112**, shown in FIG. 1). Media output component **506** is any component capable of conveying information to user **508**. For example, media output component **506** may be a display component configured to display token request interface(s). In some embodiments, media output component **506** includes an output adapter such as a video adapter and/or an audio adapter. An output adapter is operatively coupled to processor **502** and operatively connectable to an output device such as a display device, a liquid crystal display (LCD), organic light emitting diode (OLED) display, or “electronic ink” display, or an audio output device, a speaker or headphones.

[0073] In some embodiments, user system **500** includes an input device **510** for receiving input from user **508**. Input device **510** may include, for example, a keyboard, a pointing device, a mouse, a stylus, a touch sensitive panel, a touch pad, a touch screen, a gyroscope, an accelerometer, a position detector, or an audio input device. A single component such as a touch screen may function as both an output device of media output component **506** and input device **510**.

[0074] Stored in memory area **504** are, for example, computer readable instructions for providing a user interface to user **508** via media output component **506** and receiving and processing input from input device **510**. A user interface may include, among other possibilities, a web browser and client application (“app”). Web browsers enable users, such as user **508**, to display and interact with media and other information typically embedded on a web page or a website.

[0075] User system **500** may also include a communication interface **512**, which is communicatively connectable to a remote device (e.g., a server system **600**, shown in FIG. 6). Communication interface **512** may include, for example, a wired or wireless network adapter or a wireless data transceiver for use with a mobile phone network, Global System for Mobile communications (GSM), 3G, or other mobile data network or Worldwide Interoperability for Microwave Access (WIMAX).

[0076] FIG. 6 shows an example configuration of a server system **600**. Server system **600** may include, but is not limited to, token provider **108**, secure manager **104**, and/or computing device(s) associated with any party to data security computing system **100** (all shown in FIG. 1).

[0077] Server system **600** includes a processor **602** for executing instructions. Instructions may be stored in a memory area **604**, for example. Processor **602** may include one or more processing units (e.g., in a multi-core configuration) for executing instructions. The instructions may be executed within a variety of different operating systems on server system **600**, such as UNIX, LINUX, Microsoft Windows®, etc. More specifically, the instructions may cause various data manipulations on data stored in memory **604** and/or in a storage device **606** (e.g., create, read, update, and delete procedures). It should also be appreciated that upon initiation of a computer-based method, various instructions may be executed during initialization. Some operations may be required in order to perform one or more processes described herein, while other operations may be more general and/or specific to a particular programming language (e.g., C, C#, C++, Java, or other suitable programming languages, etc.).

[0078] Processor **602** is operatively coupled to a communication interface **608** such that server system **600** is capable of communicating with a remote device such as a user system **500** (shown in FIG. 5) or another server system **600**. Processor **602** may also be operatively coupled to storage device **606**. Storage device **606** is any computer-operated hardware suitable for storing and/or retrieving data. In some embodiments, storage device **606** is integrated in server system **600**. In other embodiments, storage device **606** is external to server system **600**. For example, server system **600** may include one or more hard disk drives as storage device **606**. In other embodiments, storage device **606** is external to server system **600** and may be accessed by a plurality of server systems **600**. For example, storage device **606** may include multiple storage units such as hard disks or solid state disks in a redundant array of inexpensive disks (RAID) configuration. Storage device **606** may include a storage area network (SAN) and/or a network attached storage (NAS) system.

[0079] In some embodiments, processor **602** is operatively coupled to storage device **606** via a storage interface **610**. Storage interface **610** is any component capable of providing processor **602** with access to storage device **606**. Storage interface **610** may include, for example, an Advanced Technology Attachment (ATA) adapter, a Serial ATA (SATA) adapter, a Small Computer System Interface (SCSI) adapter, a RAID controller, a SAN adapter, a network adapter, and/or any component providing processor **602** with access to storage device **606**.

[0080] Memory area **604** may include, but are not limited to, random access memory (RAM) such as dynamic RAM (DRAM) or static RAM (SRAM), read-only memory (ROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), and non-volatile RAM (NVRAM). The above memory types are exemplary only, and are thus not limiting as to the types of memory usable for storage of a computer program.

[0081] FIG. 7 is an example flow diagram illustrating a method **700** for enabling tokenized access to sensitive user data (e.g., PII). Method **700** may be implemented using one or more components of data security computing system **100** (shown in FIG. 1). Specifically, method **700** is implemented by token provider **108** (also shown in FIG. 1).

[0082] Method **700** includes initiating **702** a secure connection with a remote client computing device (e.g., client computing device **110**) of a first data subject (e.g., data subject **112**, both shown in FIG. 1), and receiving **704**, from the remote client computing device of the first data subject, a request for an access token to provide a service provider computing device (e.g., service provider **102**, also shown in FIG. 1) with access to sensitive data associated with the first data subject. The request includes a data definition of the sensitive data to which access is to be provided and one or more authorization parameters.

[0083] Method **700** also includes generating **706** the access token that enables access to the defined sensitive data according to the one or more authorization parameters, and storing **708** the access token in a token database with the data definition and the one or more authorization parameters. Method **700** further includes transmitting **710**, to the remote client computing device of the first data subject, a response including the access token and instructions that enable the remote computing device to at least one of display the access token to the first data subject and transmit the access token to the service provider computing device.

[0084] Method **700** may include additional, fewer, and/or alternative steps. For example, in some embodiments, method **700** further includes authenticating the request for the access token. In some such embodiments, the request for the access token further includes authentication credentials input by the first data subject to the remote client computing device during a log-in process, and the authenticating includes processing the authentication credentials received from the remote client computing device. In other such embodiments, the authenticating includes transmitting an authentication request message to the remote client computing device, the authentication request message including instructions that cause the remote client computing device to prompt the first data subject to input one or more authentication credentials into the remote client computing

device, receiving, from the remote client computing device, an authentication response message including the one or more input authentication credentials, and processing the input authentication credentials.

[0085] In some embodiments, method **700** further includes receiving a token validation request message from the service provider computing device, the token validation request message including the access token and a subject identifier associated with the data subject, performing a lookup operation using at least one of the access token or the subject identifier, validating the access token when the lookup operation returns a valid and active stored access token, and transmitting a token validation response message to the service provider computing device, the token validation response message including an indication that the access token was successfully validated.

[0086] In certain embodiments, method **700** includes receiving a token validation request message from a secure manager that stores the sensitive data to which access is to be provided, the token validation request message including the access token and a subject identifier associated with the data subject, performing a lookup operation using at least one of the access token or the subject identifier, validating the access token when the lookup operation returns a valid and active stored access token, and transmitting a token validation response message to the secure manager, the token validation response message including an indication that the access token was successfully validated. In some such embodiments, the token validation response message further includes instructions that cause the secure manager to transmit the sensitive data to the service provider computing device.

[0087] In some embodiments, method **700** includes receiving, from the remote client computing device, data subject input indicating a revocation of the access token, and, in response to receiving the data subject input, deleting the stored access token or disabling the stored access token to prevent further access to the sensitive data by the service provider computing device.

[0088] In still other embodiments, the one or more authorization parameters include a validity date after which access to the sensitive data is revoked, and method **700** includes, upon reaching the validity date, deleting the stored access token or disabling the stored access token to prevent further access to the sensitive data by the service provider computing device.

[0089] As will be appreciated based on the foregoing specification, the above-described embodiments of the disclosure may be implemented using computer programming or engineering techniques including computer software, firmware, hardware or any combination or subset thereof, wherein the technical effect is to enable the use of access tokens to maintain security of PII while authorizing access to such data by one or more service providers. Any such resulting program, having computer-readable code means, may be embodied or provided within one or more computer-readable media, thereby making a computer program product, (i.e., an article of manufacture), according to the discussed embodiments of the disclosure. The computer-readable media may be, for example, but is not limited to, a fixed (hard) drive, diskette, optical disk, magnetic tape, semiconductor memory such as read-only memory (ROM), and/or any transmitting/receiving medium such as the Internet or other communication network or link. The article of manufacture containing the computer code may be made and/or used by executing the code directly from one medium, by copying the code from one medium to another medium, or by transmitting the code over a network.

[0090] These computer programs (also known as programs, software, software applications, “apps”, or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms “machine-readable medium” “computer-readable medium” refers to any computer program product, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that

receives machine instructions as a machine-readable signal. The “machine-readable medium” and “computer-readable medium,” however, do not include transitory signals. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor.

[0091] This written description uses examples to disclose the disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A data security system for enabling tokenized access to sensitive data, the data security system comprising a token provisioning computing device including a processor communicatively coupled to a memory, the token provisioning computing device configured to: receive, from a service provider computing device, a request for access to sensitive data of an individual stored in memory; transmit an authorization request message to a remote client computing device associated with the individual, the authorization request message prompting the individual to authorize provisioning of an access token to the service provider, the access token being configurable by (i) associating a data definition of one or more sensitive data elements to the access token for allowing access to be granted, and (ii) identifying one or more authorization parameters; receive, from the remote computing device, an authorization response including the data definition and the one or more authorization parameters; generate the access token that enables access to the one or more sensitive data elements according to the one or more authorization parameters; and transmit, to the service provider computing device, a response including the access token.
2. The data security system of claim 1, wherein token provisioning computing device is further configured to store the access token in a token database with the data definition and the one or more authorization parameters.
3. The data security system of claim 2, wherein the access token is further stored in the token database with an identifier of the individual, and wherein token provisioning computing device is further configured to: receive, from the service provider computing device, a token validation request message including the access token; in response to receiving the token validation request message, perform a lookup operation using at least one of the access token or the identifier of the individual; when the lookup operation returns a valid and active stored access token, validate the access token; and transmit a token validation response message to the service provider computing device, the token validation response message including an indication that the access token was successfully validated.
4. The data security system of claim 3, further comprising a secure manager integrated with or separate from the token provisioning computing device, wherein the secure manager is configured to: receive, from the service provider computing device or the token provisioning computing device, the validation response message; and provide, to the service provider computing device, access to the sensitive data elements defined by the data definition according to the one or more authorization parameters.
5. The data security system of claim 1, wherein the authorization request message includes an authentication request, requesting input of authentication credentials by the individual to the remote computing device, and wherein the authorization response message includes input authentication credentials, wherein the token provisioning computing device is further configured to process the input authentication credentials.

- 6.** The data security system of claim 1, further comprising a secure manager integrated with or separate from the token provisioning computing device, wherein the secure manager is configured to: receive, from the service provider computing device or the token provisioning computing device, the access token; and provide, to the service provider computing device, access to the sensitive data elements defined by the data definition according to the one or more authorization parameters of the access token.
- 7.** The data security system of claim 6, wherein the token provisioning computing device is further configured to: store the access token in a token database with the data definition and the one or more authorization parameters. at a future time, receive, from the service provider computing device, a data access request including an inputted access token; compare the inputted access token to the access token stored in the token database; and when the inputted access token matches the stored access token, facilitate access, to the service provider computing device, to the sensitive data elements defined by the data definition according to the one or more authorization parameters of the access token.
- 8.** The data security system of claim 1, wherein the token provisioning computing device is further configured to: receive, from the remote client computing device, user input from the individual indicating a revocation of the access token; and in response to receiving the user input, at least one of delete the stored access token or disable the stored access token to prevent further access to the sensitive data by the service provider computing device.
- 9.** The data security system of claim 1, wherein the one or more authorization parameters include a validity date after which access to the sensitive data is revoked, and wherein the token provisioning computing device is further configured to: upon reaching the validity date, at least one of delete the stored access token or disable the stored access token to prevent further access to the sensitive data by the service provider computing device.
- 10.** The data security system of claim 1, wherein the token provisioning computing device is further configured to implement the authorization request by: communicatively connecting to the remote client computing device over a secure communication channel; receiving, upon communicatively connecting to the remote client computing device, an identifier associated with individual; and accessing one or more data sources using the identifier to identify any sensitive data elements associated with the individual that are respectively stored at and available from the one or more data sources; and generating a selection form based on all sensitive data elements available from the one or more data sources.
- 11.** The data security system of claim 1, wherein the one or more authorization parameters include at least one of a validity time/date parameter, a data source parameter, or an authorized service provider parameter.
- 12.** The data security system of claim 1, wherein the data definition includes a subset of a plurality of data elements available to provide to the service provider computing device from one or more data sources.
- 13.** The data security system of claim 1, wherein the access token is one of alphanumeric code, a bar code, and a QR code.
- 14.** A computer-implemented method for enabling tokenized access to sensitive data, the method implemented using a data security system including a token provisioning computing device including a processor communicatively coupled to a memory device, the method comprising: receiving, from a service provider computing device, a request for access to sensitive data of an individual stored in memory; transmitting an authorization request message to a remote client computing device associated with the individual, the authorization request message prompting the individual to authorize provisioning of an access token to the service provider, the access token being configurable by (i) associating a data definition of one or more sensitive data elements to the access token for allowing access to be granted, and (ii) identifying one or more authorization parameters; receiving, from the remote computing device, an authorization response including the

data definition and the one or more authorization parameters; generating the access token that enables access to the one or more sensitive data elements according to the one or more authorization parameters; and transmitting, to the service provider computing device, a response including the access token.

15. The computer-implemented method of claim 14, further comprising: storing the access token in a token database with the data definition, the one or more authorization parameters, and an identifier of the individual; receiving, from the service provider computing device, a token validation request message including the access token; in response to receiving the token validation request message, performing a lookup operation using at least one of the access token or the identifier of the individual; when the lookup operation returns a valid and active stored access token, validating the access token; and transmitting a token validation response message to the service provider computing device, the token validation response message including an indication that the access token was successfully validated.

16. The computer-implemented method of claim 15, wherein the data security system further includes a secure manager integrated with or separate from the token provisioning computing device, the method further comprising: receiving, by the secure manager from the service provider computing device or the token provisioning computing device, the validation response message; and providing, by the secure manager to the service provider computing device, access to the sensitive data elements defined by the data definition according to the one or more authorization parameters.

17. The computer-implemented method of claim 14, further comprising: receiving, from the remote client computing device, user input from the individual indicating a revocation of the access token; and in response to receiving the user input, at least one of deleting the stored access token or disabling the stored access token to prevent further access to the sensitive data by the service provider computing device.

18. The computer-implemented method of claim 14, wherein the one or more authorization parameters include a validity date after which access to the sensitive data is revoked, and wherein the method further comprises: upon reaching the validity date, at least one of deleting the stored access token or disabling the stored access token to prevent further access to the sensitive data by the service provider computing device.

19. A non-transitory computer-readable storage medium having computer-executable instructions stored thereon, wherein when executed by a processor of a token provisioning computing device of a data security computing system, the computer-executable instructions cause the processor to: receive, from a service provider computing device, a request for access to sensitive data of an individual stored in memory; transmit an authorization request message to a remote client computing device associated with the individual, the authorization request message prompting the individual to authorize provisioning of an access token to the service provider, the access token being configurable by (i) associating a data definition of one or more sensitive data elements to the access token for allowing access to be granted, and (ii) identifying one or more authorization parameters; receive, from the remote computing device, an authorization response including the data definition and the one or more authorization parameters; generate the access token that enables access to the one or more sensitive data elements according to the one or more authorization parameters; and transmit, to the service provider computing device, a response including the access token.

20. The non-transitory computer-readable storage medium of claim 19, wherein the one or more authorization parameters include at least one of a validity time/date parameter or an authorized service provider parameter.
